UNIVERSITE DE BOUMERDES UNIVERSITY OF BOUMERDES	*IGEE* *University M'Hamed* *Bougara of Boumerdès*	***Last Name:*** ..... ***First Name:*** ..... ***Group:***	***Grade :***

EE321: Computer Architecture
Control (1h)
2022 – 2023

Notes: Answer briefly and clearly using the provided space. No extra sheet will be accepted.

1. Simple CPU Function and Structure (15 pts)

Consider a Little-Endian simple 8-bit CPU with the following specifications: 16-bit address bus, the memory is byte addressable, the instruction list and opcodes are given in Table 1, the instruction formats are given in Table 2. The CPU has 8 data registers (R0 to R7) and 8 address registers (A0 to A7), the register codes are given in Table 3. A0 is the program counter and A7 is the stack pointer. The number of clock cycles for each basic cycle are given in Table 4.

1.1 Give the instructions that are using: a) direct addressing, and b) indirect addressing.

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1.2 Give the size in bits of the registers: R0-R7, A0-A7, MAR, and IR. Justify.

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1.3 Give the size in bytes of the instructions: a) LDD Rd,(Y), b) INC Ad, and c) LDR Ad,(As). Justify.

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1.4 What is the difference between the two instructions: LDR Rd,(As) and LDR Ad,(As). How the CPU makes the difference between the two instructions.

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1.5 Using the same format as in Table 1, give the description of the instructions: a) POP Rd, and b) POP Ad.

1.6 Assuming all the registers are connected together using internal bata bus, give the micro-operations of the **execute** cycle of the instructions: a) MOV Ad,Y, and b) LDR Rd,(As).

1.7 Give the HEX code of the instructions: a) ADI R1,10H, b) ADI A1,78H, and c) INC A1.

1.8 Give the HEX content of the registers A0 (PC), A1, MAR, and IR after the execution of the instruction LDD A1,(F122H) stored at address 89A8H.

1.9 Which of the following states: IAC, IF, IOD, OAC, OF, DO, and OS, are involved in the instructions: a) PUSH Ad, and b) POP Ad?

1.10 Give the number of clock cycles it takes to execute (i.e. fetch and execute) the instructions: a) LDR Rd, (As), b) LDD Ad,(Y), and c) PUSH Rd . Justify.

2. Basic Concepts (5 pts)

2.1 Consider the 2's complement representation using 16 bits, complete the following equalities:

a) $(7F00H)_{2'C} = (\ldots\ldots)_{10}$ $(FF00H)_{2'C} = (\ldots\ldots)_{10}$

b) $(+257)_{10} = (\ldots\ldots)_{2'C}$ $(-257)_{10} = (\ldots\ldots)_{2'C}$

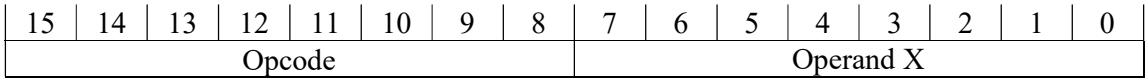
2.2 Consider a microprogrammed control unit with horizontal microinstructions. Assuming 128 control signals (internal and external), 500 microinstructions, and 3 bits for jump condition field, give the number of bits of the jump address field. Justify.

2.3 Consider a CPU with a 20-bit address bus and 16-bit data bus. Assuming the memory range ABC00H-ABDF0H of the CPU is mapped to a RAM memory chip, what is the size in bits and the number of address lines of the RAM.

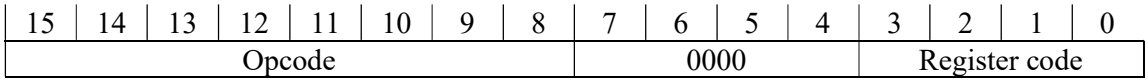
Instruction	Opcode (binary)	Instruction Size	Description
MOV Rd, X	$0001C_3C_2C_1C_0$	2 bytes	Rd = X
MOV Ad, Y	$0001C_3C_2C_1C_0$	3 bytes	Ad = Y
MOV Rd, Rs	$0010C_3C_2C_1C_0$	2 bytes	Rd = Rs
PUSH Rd	$0011C_3C_2C_1C_0$	1 byte	A7 = A7-1, [A7] = Rd
PUSH Ad	$0011C_3C_2C_1C_0$		
LDD Rd, (Y)	$0100C_3C_2C_1C_0$		Rd = [Y]
LDR Rd, (As)	$0101C_3C_2C_1C_0$		Rd = [As]
LDR Ad, (As)	$0101C_3C_2C_1C_0$		Ad = [As]
STD (Y), Rd	$0110C_3C_2C_1C_0$		[Y] = Rd
STD (Ad), Rs	$0111C_3C_2C_1C_0$		[Ad] = Rs
ADI Rd, X	$1000C_3C_2C_1C_0$		Rd = Rd + X
ADI Ad, Y	$1000C_3C_2C_1C_0$		Ad = An + Y
ADD Rd, (As)	$1001C_3C_2C_1C_0$		Rd = Rd+[As]
ADR Rd, Rs ADR Ad, As	$1010C_3C_2C_1C_0$		Rd = Rd + Rs Ad = Ad + As
POP Rd POP Ad	$1011C_3C_2C_1C_0$		
INC Rd INC Ad	$1100C_3C_2C_1C_0$		Rd = Rd + 1 Ad = Ad + 1
HLT	1111000		Stop the execution of the program

Rd =Data register (it can be any data register).
Rs = Data register (it can be any data register).
Ad = Address register (it can be any address register).
As = Address register (it can be any address register).
$C_3C_2C_1C_0$ = Binary code of the operand (Rd / Ad) (see Table 3).
X =8-bit 2's complement value.
Y =16-bit 2's complement value.
[…] = Content of memory location.

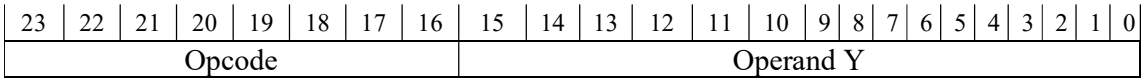
Table 1: Instructions description.



2-bytes instruction format.



2-bytes instruction format.



3-bytes instruction format

Table 2: 2-bytes and 3-bytes instructions formats.

Register	R0	R1	R2	R3	R4	R5	R6	R7
Code	0000	0001	0010	0011	0100	0101	0110	0111

Register	A0	A1	A2	A3	A4	A5	A6	A7
Code	1000	1001	1010	1011	1100	1101	1110	1111

Table 3: Binary codes of data and address registers.

Cycle	Number of clock cycles
Opcode fetch	4 clock cycles
1-byte operand fetch	3 clock cycles
Memory read	3 clock cycles
Memory write	3 clock cycles
Do operation	1 clock cycle

Table 4: Number of clock cycle for each basic cycle.